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Current Development in the Synthesis of Benzimidazole-Quinoline Hybrid Analogues and their Biological Applications

Zebabanu Khalifa, Rachana Upadhyay, Premlata Kumari and Amit B. Patel

Abstract

Heterocyclic compounds have an essential role in many domains of medicinal chemistry. Many pharmaceutical industries use and investigate nitrogen-containing heterocycles because they are crucial in discovering and developing novel therapeutically active compounds. The benzimidazole moiety is a fundamental component of many heterocyclic scaffolds, which play an important role in producing a wide range of biological activities. Similarly, quinoline is also a versatile bicyclic heterocyclic scaffold with many medicinal applications. It is an essential scaffold for drug discovery leads, and it plays a significant role in medicinal chemistry and has biological activities similar to benzimidazole scaffolds. The present chapter discusses the quinoline-benzimidazole hybrids scaffolds and their potential pharmacological activities.

Keywords: quinoline, benzimidazole, hybrid molecules, synthesis, pharmacological activity

1. Introduction

Heterocycles are common elements found in the majority of commercial drugs and are a target of medicinal chemistry in the drug delivery process [1]. In recent years, nitrogen-containing five- and six-member heterocycles have received a significant amount of interest because of their major pharmacological and synthetic implications [2]. Structure elucidation, identification of novel biologically active compounds, development of efficient therapies and discovering the mode of action of newer molecules are all critical aspects of medicinal chemistry that nitrogen-containing heterocyclic compounds perform. They are commonly employed in drug development because of the discovery of robust synthetic pathways that can rapidly produce large amounts of desired chemicals, which helps to accelerate the drug development process. Their pharmacophores have a wide range of pharmacological activities against disease or disorder [3, 4].

The “one drug, one target, one disease” strategy is no longer appropriate in today’s complicated and infectious diseases [5]. Drug resistance is a problem that cannot be solved using usual disease treatment methods. The hybridisation of physiologically active compounds is a potent drug discovery approach that can be utilised to treat a wide range of diseases. It opens up the possibility of better medications [6]. Benzimidazole is the building block of various synthetic medicinal and biochemical compounds with important biological activities like anticancer [7, 8], antimicrobial [9, 10], antibacterial [11, 12], antimycobacterial [13, 14], anti-inflammatory [15, 16], etc. Similarly, quinoline is found in various natural products. It is frequently utilised to explore a range of bioactive compounds with varying pharmacological properties like antifungal [17, 18], anti-inflammatory [19, 20], anticancer [21, 22], antimicrobial [23, 24], anticoagulant [25, 26], antiviral [27, 28], antimalarial [29, 30], antitrypanosomal [30, 31] etc. Due to the therapeutic efficacy of benzimidazole and quinoline scaffolds, medicinal chemists have been interested in developing hybrid analogues with enhanced potency by incorporating them, and molecular hybridisation technology has been employed to do so. With a virtual planning and developing procedure, finding the enhanced pharmacological activity of new or modified present drugs is quite challenging. In such cases, these benzimidazole-quinoline hybrid compounds play a crucial role since they can reduce time and expenses by employing techniques including X-ray screening, molecular docking, NMR skeletons of biomolecules and computer-aided drug design [32, 33].

The fundamental purpose of benzimidazole-quinoline hybrid compounds is to strengthen their potential to interact with several biological targets. Due to their excellent potential in drug discovery and development, we have introduced this study of the pharmacological activities of benzimidazole-incorporated quinoline compounds.

2. Biological applications of benzimidazole-quinoline hybrids

2.1 Anticancer activity

Hranjec and co-workers developed the synthesis and biological characterisation of novel benzimidazo[1,2-*a*]quinoline-6-carbonitriles derivatives (**Figure 1**). The reaction was initiated by condensing 2-cyanomethylbenzimidazole with heteroaromatic aldehyde yielded acrylic substituted acrylonitriles. The photochemical dehydrocyclisation procedure was utilised to produce benzimidazo[1,2-*a*]quinoline-6-carbonitriles and heteroaromatic fluorenes. All the synthesised compounds were investigated for antiproliferative efficacy against five tumour cell lines such as HeLa, MiaPaCa-2, SW

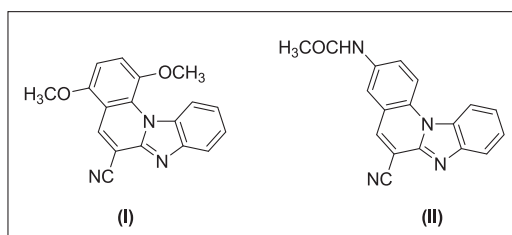


Figure 1.
Benzo[4,5]imidazo[1,2-*a*]quinoline-6-carbonitrile methoxy/anilide derivatives.

620, MCF-7 and H460. Among them, compounds (I) and (II) showed good antiproliferative activity against all the cell lines with an IC₅₀ range of 2–90 µM [34].

Perin *et al.* synthesised benzimidazo[1,2-*a*]quinolines and benzimidazo[1,2-*a*]quinolines-6-carbonitriles derivatives by the thermal reaction using sulfolane for dehydrohalogenation cyclisation at 280°C followed by UV/Vis spectroscopy (Figure 2). Out of all the synthesised compounds (III), (IV), (V), (VI), (VII) and (VIII) showed strong antiproliferative activity with IC₅₀ range of 2–19 µM [35].

Further, Perin and co-workers have also synthesised amino-substituted cyclic benzimidazo[1,2-*a*]quinoline derivatives and tested *in vitro* for their antiproliferative, antibacterial and antiviral activities (Figure 3). Low to moderate yields of amide substituted cyclic derivatives were obtained by condensing acyl halides with an excess quantity of the corresponding amines in absolute dichloromethane. The antiproliferative activity of (IX), (X) and (XI) was found to be most promising with an IC₅₀ value ranging between 0.48–4.1 µM against HeLa cell line and 0.24–0.69 µM against MCF7 cell line, and it also caused apoptosis in human cervical carcinoma HeLa cells at micromolar doses. Antiviral activity was observed in compounds (XII), (XIII) and (XIV) having a side chain at position-2 against the herpes simplex virus (HSV) with an EC₅₀ value of 1.8–6.8 µM and human coronavirus with EC₅₀ value of 4–12 µM. In addition, (IX) and (X) substituting an amide side chain position-6 of the tetracyclic skeleton were efficacious against *S. epidermidis* and *C. albicans* [36].

Perin *et al.* synthesised 2-substituted benzimidazo[1,2-*a*]-quinoline derivatives by microwave-assisted amination of 2-chloro/fluorobenzo[4,5]imidazo[1,2-*a*]quinoline-6-carbonitrile precursors, which were prepared by photochemical dehydrogenation of non-fused *E*-2-(2-benzimidazolyl)-3-(4-halophenyl)acrylonitriles with 4-halobenzaldehyde (Figure 4). Furthermore, the antiproliferative efficacy of amino-substituted derivatives was investigated. Compound (XV) showed prominent activity against HCT116 and H460 cells with IC₅₀ values of 0.5 and 0.3 µM, respectively. The compounds (XVI) and (XVII) with the presence of cyclic tertiary amino substituent displayed excellent antiproliferative action against HCT116 and H460 cell lines with IC₅₀ range of 0.06–0.3 µM [37].

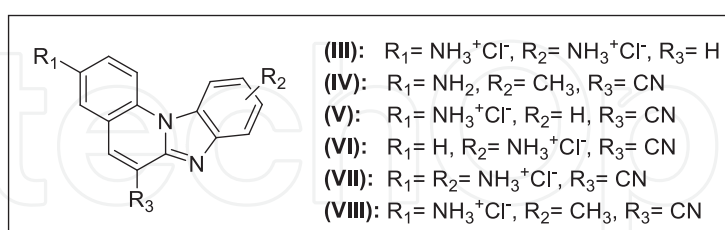


Figure 2.
Amino substituted benzo[4,5]imidazo[1,2-*a*]quinoline derivatives.

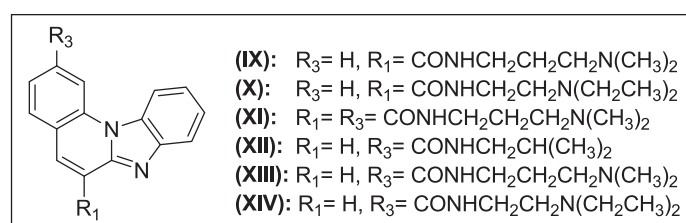


Figure 3.
Benzo[4,5]imidazo[1,2-*a*]quinoline amide derivatives.

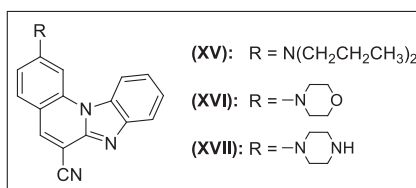


Figure 4.
 Benzo[4,5]imidazo[1,2-a]quinoline-6-carbonitrile derivatives.

Tantawy *et al.* synthesised novel 2-(1*H*-benzo[*d*]imidazol-2-yl)quinoline-4-carboxylic acid derivatives by the reaction of 2-acetylimidazoles with isatin (**Figure 5**). Compounds (**XVIII**), (**XIX**) and (**XX**) displayed excellent *in vitro* antitumour activity with IC_{50} in the range of 12.7–16.13 $\mu\text{g/mL}$ (percentage inhibition of 71.9–74.0%) against MCF7 breast cancer cell lines [38].

Shi and co-workers have designed and synthesised *N*-(2-phenyl-1*H*-benzo[*d*]imidazol-5-yl) quinolin-4-amine derivatives as potent inhibitors of VEGFR-2 (KDR) kinase (**Figure 6**). These derivatives were prepared by the condensation of 4-chloroquinoline with benzimidazole quinoline-4-amines. The synthesised compound (**XXI**) showed the most potent inhibitory activity against VEGFR-2 with IC_{50} value of 0.03 μM . Moreover, this compound also showed the highest antiproliferative activity against MCF-7 and Hep-G2 cell lines with IC_{50} values of 1.2 and 13.3 μM , respectively [39].

Brajsa *et al.* prepared 2-Imidazoliny-substituted triaza-benzo[*c*]fluorenes derivatives from triaza-benzo[*c*]fluorenes using the Pinner reaction and tested for antitumour efficacy in 2D and 3D cell culture (**Figure 7**). The compound (**XXII**) showed the most prominent activity against PANC-1 and MDA-MB-231 cell lines with IC_{50} values of 1 and 0.2 μM , respectively [40].

Kuang *et al.* prepared a series of 3-(1*H*-benzimidazol-2-yl)quinoline-2(1*H*)-one analogues by coupling reaction of 2-oxo-quinoline-3-carbaldehyde with various *o*-phenylenediamines (**Figure 8**). The HepG2 tumour growth was effectively inhibited by (**XXIII**) with IC_{50} value of 8.45 μM in an *in vitro* antitumour experiment. In

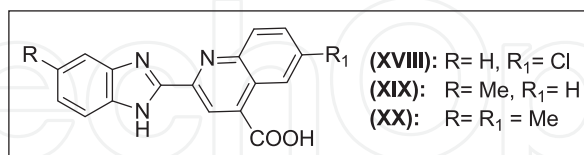


Figure 5.
 2-(1*H*-benzo[*d*]imidazol-2-yl)quinoline-4-carboxylic acid derivatives.

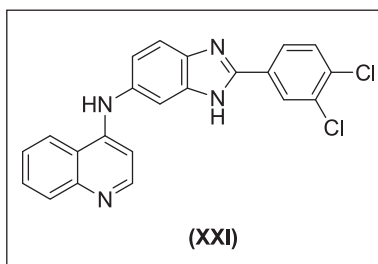


Figure 6.
N-(2-(3,4-dichlorophenyl)-1*H*-benzo[*d*]imidazol-6-yl)quinolin-4-amine derivatives.

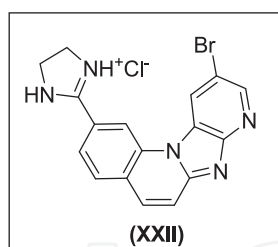


Figure 7.
2-(10-Bromopyrido[2',3'4,5]imidazo[1,2-a]quinoline-2-yl)-4,5-dihydro-1H-imidazol-3-ium chloride derivatives.

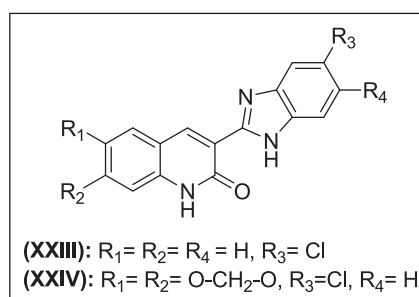


Figure 8.
3-(1H-benzo[d]imidazol-2-yl)quinolin-2(1H)-one derivatives.

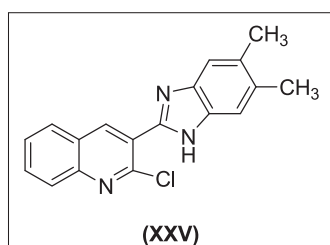


Figure 9.
2-Chloro-3-(5,6-dimethyl-1H-benzo[d]imidazol-2-yl)quinoline derivatives.

addition, the compound **(XXIV)** showed excellent antiproliferative activity against BEL-7402 with IC_{50} value of $9.06 \mu M$ [41].

Similarly, Kuang *et al.* further synthesised 2-chloro-3-(1H-benzo[d]imidazol-2-yl)quinoline derivatives as potent antitumour agents (**Figure 9**). *In vitro* antitumour assay findings revealed that some compounds had moderate to high inhibitory effects against HepG2, SK-OV-3, NCI-H460 and BEL-7404 tumour cells, as compared to 5-FU and cisplatin. In the HepG2 xenograft model, compound **(XXV)** effectively inhibited tumour growth with IC_{50} value $7.54 \mu M$ [42].

Macan and co-workers prepared 2-Fluoro-5-(1H-1,2,3-triazol-1-yl)benzo[4,5]imidazo[1,2-a]quinoline-6-carbonitrile derivatives *via* Huisgen 1,3-dipolar cycloaddition of the azide derivatives and the corresponding alkynes using $Cu(OAc)_2$ as a catalyst (**Figure 10**). The presence of 3-chloropropyl functional group in compound **(XXVI)** and 2-hydroxyethyl side-chain in compound **(XXVII)** exhibited the most pronounced growth-inhibitory effect on colon cancer (HCT116) cells with IC_{50} value of $0.5 \mu M$ and $0.6 \mu M$, respectively [43].

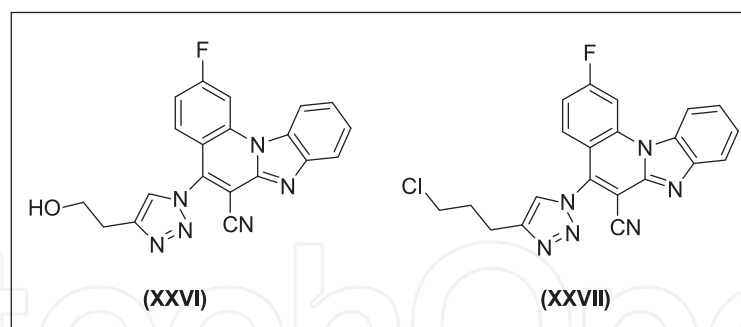


Figure 10.
2-Fluoro-5-(1H-1,2,3-triazol-1-yl)benzo[4,5]imidazo[1,2-a]quinoline-6-carbonitrile derivatives.

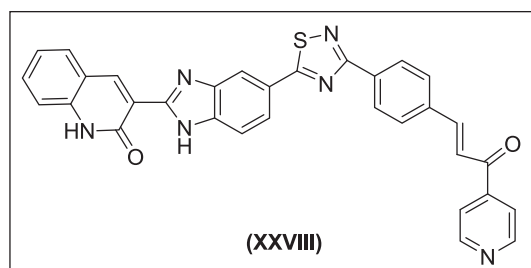


Figure 11.
(E)-3-(5-(3-(4-(3-oxo-3-(pyridin-4-yl)prop-1-en-1-yl)phenyl)-1,2,4-thiadiazol-5-yl)-1H-benzo[d]imidazol-2-yl)quinoline-2(1H)-one derivatives.

Pragathi and co-workers have synthesised a series of chalcone incorporated 2-quinoline-benzimidazole-1,2,4-thiadiazoles by aldol condensation (**Figure 11**). Synthesised compounds were examined for their anticancer activities against different cancer cell lines such as MCF-7, A549, Colo-205 and A2780. From all the investigated compounds, (**XXVIII**) showed excellent anticancer activities with IC_{50} range of 0.012–1.46 μ M [44].

Gaikwad *et al.* developed a series of 1,2,3-triazole-based quinoline-benzimidazole hybrid scaffolds by a click reaction using the [3+2] azide-alkyne cycloaddition reaction using copper(I)Iodide in DMF (**Figure 12**). Synthesised compounds were screened against a panel of NCI-60 human cancer cell lines for in vitro cytotoxicity evaluation. The compound (**XXIX**) showed an excellent antiproliferative effect with a significant IC_{50} value of 0.59 μ M against the human breast cancer BT-474 cell line [45].

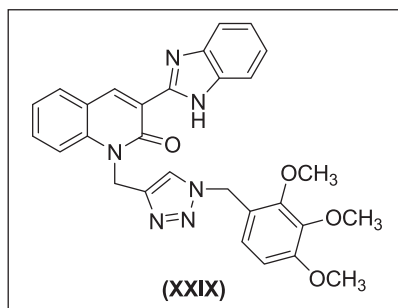


Figure 12.
3-(1H-benzo[d]imidazol-2-yl)-1-((1-(2,3,4-trimethoxybenzyl)-1H-1,2,3-triazol-4-yl)methyl)quinolin-2(1H)-one derivatives.

2.2 Antimicrobial activity

Sonar *et al.* synthesised 4-((5-difluoromethoxy)-1H-benzo[d]imidazol-2-ylthio) methyl)tetrazolo[1,5-a]quinolines derivatives (**Figure 13**). These targeted molecules were synthesised by the conversion of 2-chloroquinoline-3-carbaldehyde to tetrazolo [1,5-a]quinoline-4-carbaldehyde after treatment with sodium azide, followed by reduction to the corresponding alcohol derivatives and conversion to chloride with thionyl chloride and coupling with 5-(difluoromethoxy)-1H-benzo[d]-imidazol-2-thiol. The compounds (**XXX**), (**XXXI**), (**XXXII**), (**XXXIII**) and (**XXXIV**) were evaluated for their antibacterial activity against Gram-positive (*B. subtilis* and *S. aureus*) and Gram-negative (*E. Coli* and *S. aboney*) microorganisms. The synthesised compounds demonstrated significant activity against all the bacteria with zone of inhibition ranging from 7 to 17 mm at MIC level of 10 mg/mL [46].

Chaudhari *et al.* synthesised 8-(1-alkyl/alkylsulphonyl/alkoxycarbonyl-benzimidazol-2ylmethoxy)-5-chloroquinoline derivatives (**Figure 14**). The first step was the reaction of 5-chloro-8-hydroxyquinoline with monochloroacetic acid methyl ester followed by hydrolysis to form 5-chloroquinolin-8-yl)oxy]acetic acid. Which was further reacted with *o*-phenylenediamine to obtain 8-[(1H-benzimidazol-2-yl) methoxy]-5-chloroquinoline. The final analogues were synthesised by the reaction of 8-[(1H-benzimidazol-2-yl) methoxy]-5-chloroquinoline with different electrophiles in the presence of an appropriate base. Synthesised compounds were tested for their antifungal activity, among them compound (**XXXV**), (**XXXVI**), (**XXXVII**) and (**XXXVIII**) showed good antifungal activity against *A. niger* MTCC282 with 18–19 mm zone of inhibition at 50 µg/mL MIC. In addition, compound (**XXXIX**) showed potent antibacterial activity with 15 mm zone of inhibition (MIC, 25 µg/mL) against *S. aureus* MTCC-96 and *S. typhimurium* MTCC-98 [47].

Gowda *et al.* used one-step method to synthesise 2-(1H-benzimidazol-2-yl)-6-substituted thieno[2,3-b]quinoline derivatives (**Figure 15**). They introduced nucleophilic substitution reaction of 2-(chloromethyl)-1H-benzimidazole 2/2-(mercaptomethyl)-1H-benzimidazoles followed by the cyclisation with 2-mercapto-quinoline-3-carbaldehyde 1/2-chloroquinoline-3-carbaldehyde. The antibacterial activity of the synthesised compounds was tested *in vitro* against Gram-positive (*E. coli* and *S. aureus*) and Gram-negative (*P. aeruginosa* and *K. pneumoniae*) pathogens.

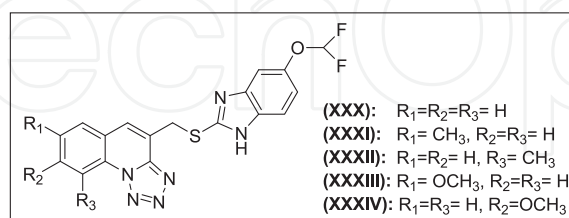


Figure 13.
4-((5-(Difluoromethoxy)-1H-benzo[d]imidazol-2-yl)thiomethyl)tetrazolo[1,5-a]quinoline derivatives.

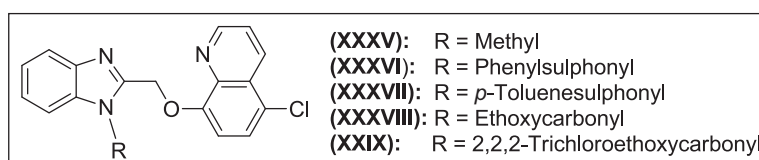
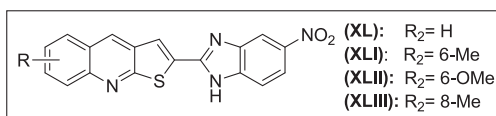


Figure 14.
8-((1H-benzo[d]imidazol-2-yl)methoxy)-5-chloroquinoline derivatives.

**Figure 15.**

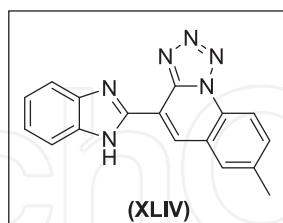
2-(1H-benzo[d]imidazo-2-yl)thieno[2,3-b]quinolines derivatives.

Compounds (XL), (XLI), (XLII) and (XLIII) with the substitution of nitro group at the position-5 of benzimidazole ring showed highest MIC at 12.5 $\mu\text{g/mL}$ [48].

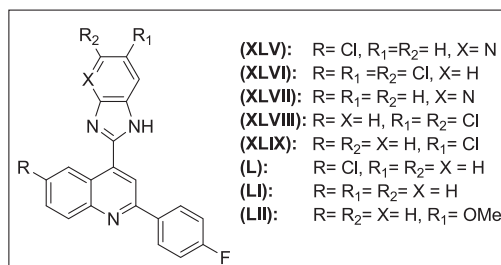
Mungra *et al.* synthesised a new class of potent quinoline fused benzimidazole derivatives (Figure 16). The final compounds were prepared by the reaction of tetrazolo[1,5-*a*]quinoline-4-carbaldehyde with *o*-phenylenediamine by microwave irradiation. Compounds were further tested for their antimicrobial activity. Among all compounds, (XLIV) displayed excellent activity against Gram-positive bacteria *B. subtilis* with MIC value of 100 $\mu\text{g/mL}$ [49].

Garudachari *et al.* synthesised quinoline incorporated benzimidazole from aniline and isatin via the multistep process (Figure 17). These derivatives were synthesised by multi-component one-pot Doebner and Pfitzinger reaction. Newly synthesised compounds were tested *in vitro* for their antibacterial and antifungal activities. From them, compounds (XLV), (XLVI), (XLVII) and (XLVIII) showed significant antibacterial activity against *S. aureus* and *E. coli* with a zone of inhibition in the range of 10–16 mm at MIC value of 6.25 $\mu\text{g/mL}$. Compound (XLIX) was showed potent antifungal activity against *A. niger*, and *Penicillium sp.* with 12–15 mm zone of inhibition, and (L), (LI) and (LII) showed good antifungal activity against *A. niger* and *Penicillium sp.* at 6.25 $\mu\text{g/mL}$ with 12–16 mm zone of inhibition [50].

De Souza *et al.* introduced well-known fluorescent benzoimidazo[1,2-*a*]quinoline derivatives with DNA binding and anticancer properties. Compounds were tested as bifunctional agents that could detect yeast biofilms on stainless steel surfaces and act as biocidal agents (Figure 18). The biocidal activity of benzimidazole compound

**Figure 16.**

4-(1H-benzo[d]imidazol-2-yl)-7-methyltetrazolo[1,5-a]quinoline derivatives.

**Figure 17.**

4-(1H-benzo[d]imidazol-2-yl)-2-(4-fluorophenyl)quinoline derivatives.

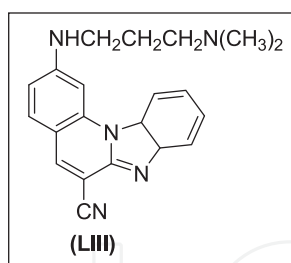


Figure 18.
2-((3-(Dimethylamino)propyl)amino)-7a,11a-dihydrobenzo[4,5]imidazo[1,2-a]quinoline-6-carbonitrile derivatives.

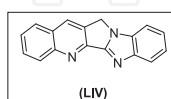


Figure 19.
12H-benzo[4',5']imidazo[1',2':1,2]pyrrolo[3,4-b]quinoline derivatives.

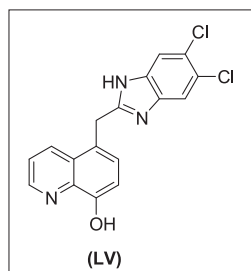


Figure 20.
5-((5,6-Dichloro-1H-benzo[d]imidazol-2-yl)methyl)quinolin-8-ol derivatives.

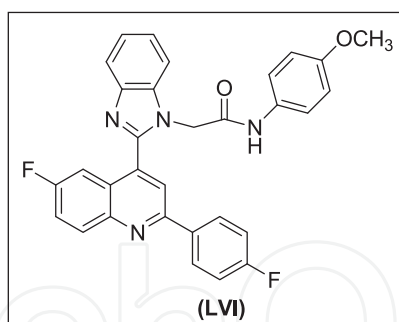
(LIII) against yeast cells and biofilms seemed promising. The compound showed significant activity against *C. Albicans* CA01 with MIC value of 4 µg/mL [51].

Villa *et al.* synthesised 12H-benzo[4',5']imidazo[1',2':1,2]pyrrolo[3,4-b]quinoline derivatives using two different pathways (Figure 19). In the first path, titled analogues were prepared by the reaction of arylimines and 20 mol% of BF₃O(C₂H₅)₂ in dichloroethane. The arylimines were obtained by the reaction of benzimidazole carbaldehyde with arylamine in the presence of InCl₃. However, another path was accomplished by a one-pot intramolecular Povarov reaction without isolating the intermediates. Among them, compound (LIV) significantly reduced the metabolic activity of *C. albicans* biofilms fungal cells at MIC value of 0.5 µg/mL [52].

El Faydy and co-workers synthesised 8-quinolinol analogues bearing a benzimidazole moiety by condensing 5-(carboxymethyl)-8-quinolinol with substituted *o*-phenylenediamines in an acidic medium (Figure 20). The synthesised compounds were tested in vitro against two Gram-positive (*B. subtilis* and *S. aureus*) and two Gram-negative (*E. Ludwigii* and *E. coli*) microorganisms. The compound (LV) showed good antibacterial activity against all the microorganisms, with 28–42 mm zone of inhibition at MIC in the range of 10–20 µg/mL [53].

2.3 Miscellaneous activity

El-Feky *et al.* investigated the anti-inflammatory potential of several novel *N*-(substituted-phenyl)-2-(2-(6-fluoro-2(4-fluorophenyl)-quinoline-4-yl)-1H-

**Figure 21.**

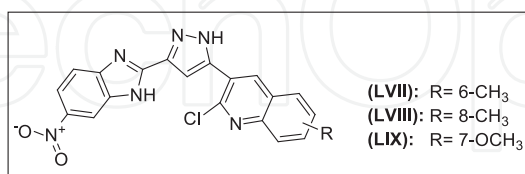
2-(2-(6-Fluoro-2-(4-fluorophenyl)quinolin-4-yl)-1H-benzo[d]imidazol-1-yl)-N-(4-methoxyphenyl)acetamide derivatives.

benzimidazol-1-yl)acetamides derivatives by the reaction with 4-(1H-benzo[d]imidazol-2-yl)-6-fluoro-2-(4-fluorophenyl)quinoline with 2-chloro-N-arylacetamide (**Figure 21**). Synthesised compounds were investigated for their anti-inflammatory activity. Among them, the compound (**LVI**) showed the highest activity with 55 mg/kg dose (%) [54].

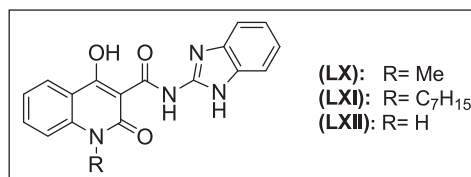
Madawali *et al.* synthesised novel 2-chloro-3-[3-(6-nitro-1H-benzimidazol-2-yl)-1H-pyrazol-5-yl]quinolines derivatives from 2-chloro-3-[3-(6-nitro-1H-benzimidazol-2-yl)-4,5-dihydro-1H-pyrazol-5-yl]quinolines by dehydrogenation reaction (**Figure 22**). Among them, compounds (**LVII**), (**LVIII**) and (**LIX**) showed excellent anti-inflammatory activity with excellent % inhibition of denaturation 82.81, 82.35 and 82.81, respectively [55].

Ukrainets and co-workers synthesised benzimidazo-2-ylamides of 4-hydroxy-2-oxo-1,2-dihydroquinoline-3-carboxylic acids (**Figure 23**). The targeted compounds were prepared by the well-known reaction, which involved heating ethyl esters with 2-aminobenzimidazole. They were tested for their antituberculosis and antithyroid activities. Compounds (**LX**) and (**LXI**) showed the highest activity against *M. tuberculosis* H37Rv ATCC 27294 with MIC value of 1.56 µg/mL. Moreover, the compound (**LXII**) (1.46 nmol/l) showed higher antithyroid activity as compared to the standard drug Mercazolyl (1.63 nmol/l) against the triiodothyronine (T3) hormone [56].

Mantu *et al.* designed and synthesised 2-(1H-benzo[d]imidazol-1-yl)-N-(quinoline-8-yl)acetamide analogues from 2-chloro-N-(quinolin-8-yl)acetamide and imidazole

**Figure 22.**

2-Chloro-3-[3-(6-nitro-1H-benzimidazol-2-yl)-4,5-dihydro-1H-pyrazol-5-yl]quinoline derivatives.

**Figure 23.**

N-(1H-benzo[d]imidazol-1-yl)-4-hydroxy-2-oxo-1,2-dihydroquinoline-3-carboxamide derivatives.

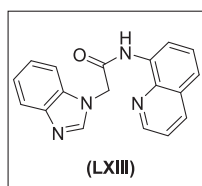


Figure 24.
2-(1H-benzo[d]imidazol-1-yl)-N-(quinoline-8-yl)acetamide derivatives.

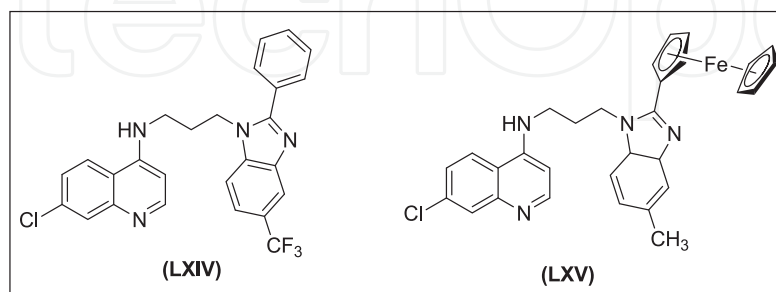


Figure 25.
7-Chloro-N-(3-(2-phenyl/ferrocenyl-5-(trifluoromethyl)-1H-benzo[d]imidazol-1-yl)propyl)quinolin-4-amine derivatives.

derivatives (**Figure 24**). The synthesised derivatives were tested *in vitro* for anticancer and antimycobacterial activities. Among all hybrid compounds, (**LXIII**) showed good antitumour activity against Renal Cancer A498 (52.92 μM) and Breast Cancer MDA-MB-468 (56.54 μM). This compound also showed antimycobacterial activity against *M. tuberculosis* H37Rv with MIC >100 μM and IC₉₀ value of 77 $\mu\text{g/mL}$ [57].

Baartzes and co-workers synthesised phenyl-benzimidazole and ferrocenyl-benzimidazoles fused aminoquinoline derivatives and evaluated their antiplasmodial activity (**Figure 25**). Out of all the synthesised hybrids analogues, phenyl (**LXIV**) and ferrocenyl (**LXV**) showed excellent antiplasmodial activity against multi-drug resistant K1 strains of the human malaria parasite at IC₅₀ values of 0.151 and 0.283 μM , respectively. They were further assessed *in vivo* against *Plasmodium berghei* infected mice. Treatment with either compound decreased parasitemia, with (**LXV**) demonstrating superior activity [58].

Pomel *et al.* synthesised 7-chloro-4-(4-(5-fluoro-1H-benzo[d]imidazol-2-yl)phenoxy)quinoline derivatives by the reaction of 2-phenol-1H-6-fluorobenzimidazole and 4,7-dichloroquinoline chloride. *T. b. gambiense*, a human parasite, was used *in vitro* to test antitrypanosomal activity (**Figure 26**). The compound (**LXVI**) showed the best *in vitro* activity with IC₅₀ value of 1.98 μM , cytotoxicity CC₅₀ value 10.56 μM and selective index (SI) was 5.3 [59].

Bharadwaj *et al.* synthesised benzimidazole-containing quinolinyl oxadiazoles from quinoline-4-carboxylic acid and benzimidazole hydrazide (**Figure 27**). These

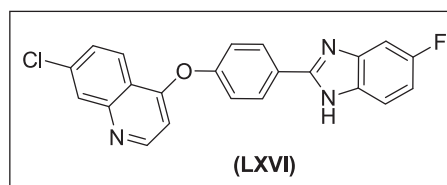


Figure 26.
7-Chloro-4-(4-(5-fluoro-1H-benzo[d]imidazol-2-yl)phenoxy)quinoline derivatives.

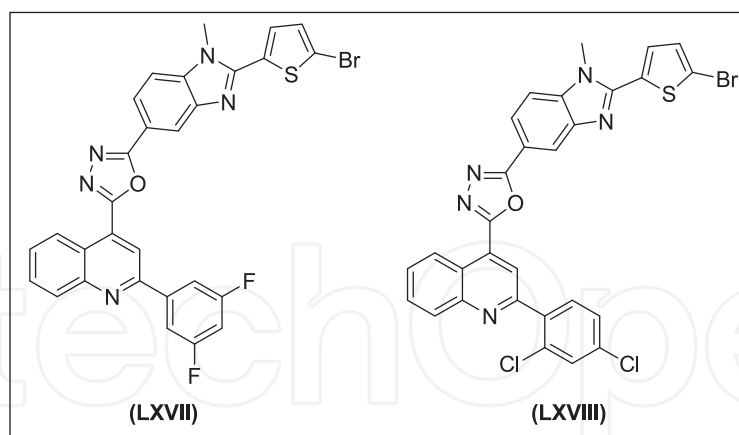


Figure 27.
2-(2-(5-Bromothiophen-2-yl)-1-methyl-1H-benzo[d]imidazol-5-yl)-5-(2-phenylquinolin-4-yl)-1,3,4-oxadiazole derivatives.

derivatives were investigated for their antidiabetic, anticoagulant and antiplatelet activity. The anticoagulant activity showed by compound (LXVII) and (LXVIII) with IC_{50} values of 0.66 and 0.68 $\mu\text{g/mL}$, respectively. The compound (LXVII) also showed antiplatelet activity [60].

2.4 Summary

The study summarises the detailed literature review on recent advancements in synthesis and the pharmacological properties of benzimidazole-integrated quinoline hybrids. The most common and efficient approach for the synthesis of benzimidazole derivatives is the condensation reaction between *o*-Phenylenediamine and various carbonyl compounds. However, quinoline derivatives are basically synthesised by the Skraup reaction. It involves a reaction with aniline or a substituted aniline with glycerol in the presence of sulphuric acid, ferrous sulfate, and nitrobenzene. The result and discussion showed that benzimidazole-quinoline hybrids could display various biological activities such as anticancer, antimicrobial, anti-inflammatory, antimycobacterial, antiplasmodial, antitrypanosomal, anticoagulant and antithyroid activities. This established a new level of revolution for the synthesis of hybrid

Entry	Type	Activity
(XVII)	Anticancer activity	$IC_{50} = 0.06\text{--}0.2 \mu\text{M}$
(LIV)	Antimicrobial activity	$MIC = 0.5 \mu\text{g/mL}$
(LVI)	Anti-inflammatory activity	Protection dose (%) = 55
(LX) and (LXI)	Antimycobacterial activity	$MIC = 1.56 \mu\text{g/mL}$
(LXIV)	Antiplasmodial activity	$IC_{50} = 0.151 \mu\text{M}$
(LXVI)	Antitrypanosomal activity	$IC_{50} = 1.98 \mu\text{M}$
(LXVII)	Anticoagulant activity	$IC_{50} = 0.66 \mu\text{g/mL}$
(LXII)	Antithyroid activity	1.46 nmol/l

Table 1.
Biological activity of benzimidazole-quinoline hybrids.

molecules. The benzimidazole-quinoline hybrids with the highest reported activities are listed in **Table 1**.

3. Conclusions

The present chapter narrates the distinct pharmacological actions of benzimidazole integrated quinoline hybrid scaffolds. The prime objective of this study is to investigate the pharmacological potential of hybrid compounds consisting of benzimidazole and quinoline moieties in a single molecule. This can help improve their potential to interact with a wide range of biological targets and enhance their pharmacological properties. The efforts are rationalised to develop such scaffolds that could impart numerous pharmacological effects to accomplish the desired aim. Further, the structure–activity relationship, pharmacokinetic, pharmacodynamic, and toxicological studies of benzimidazole-quinoline hybrid derivatives will be highly beneficial for designing and developing novel bioactive agents in the future.

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Conflict of interest

The authors declare no conflict of interest.

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